

IMAGE DENOISING USING SPATIAL FILTERING TECHNIQUES

Digital Image Processing Mini Project

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1. INTRODUCTION

Images obtained from digital cameras, sensors, scanners, and communication systems are rarely completely free from unwanted variations. These unwanted variations are commonly referred to as image noise. Noise may be introduced during image acquisition, transmission, storage, or other stages of image processing.

The presence of noise can reduce image quality by changing pixel intensities and introducing unwanted structures. In severe cases, noise can make important image details difficult to recognize. Therefore, removing or reducing noise is an important task in Digital Image Processing.

Image denoising aims to suppress unwanted noise while retaining important structures such as edges, textures, and intensity variations. Spatial filtering is one of the fundamental approaches to image denoising. In spatial filtering, the value of a pixel is modified using information from a local neighborhood around that pixel.

This project focuses on three commonly used spatial filters:

1. Mean Filter
2. Median Filter
3. Gaussian Filter

Two different types of artificial noise are considered:

1. Gaussian noise
2. Salt-and-pepper noise

The project compares the behavior of the three filters under both noise conditions. Instead of relying only on visual inspection, quantitative evaluation is performed using MSE, PSNR, and SSIM.

The experiment was implemented in Google Colab using Python. The uploaded images were resized to 512×512 pixels before processing. The final results were averaged across the three images to obtain a more consistent comparison.

2. PROBLEM STATEMENT

Digital images can contain different forms of noise that degrade their quality. A filter that performs well for one type of noise may not provide equally effective results for another type.

The problem addressed in this project is:

To investigate and compare the effectiveness of Mean, Median, and Gaussian spatial filters in reducing Gaussian noise and salt-and-pepper noise from digital images.

The project aims to determine how the choice of spatial filter affects the quantitative quality of the restored image.

3. OBJECTIVES

The main objectives of this project are:

1. To study the effect of noise on digital images.
2. To generate Gaussian noise artificially.
3. To generate salt-and-pepper noise artificially.
4. To implement Mean, Median, and Gaussian spatial filters.
5. To apply the filters to noisy images.
6. To compare the visual quality of the filtered images.
7. To evaluate the restored images using MSE, PSNR, and SSIM.
8. To perform the experiment on multiple images rather than a single image.
9. To calculate average performance for each filter and noise type.
10. To analyze how the type of noise affects filter performance.

4. DATASET AND IMAGE DESCRIPTION

4.1 Image Dataset

A small self-selected image dataset was used for the experiment. Three color images were uploaded directly to the Google Colab environment:

- img1.jpeg
- img2.jpeg

- img3.jpeg

The images were used as clean reference images. Artificial noise was subsequently added to these images so that the denoising performance could be evaluated against the known original images.

The notebook implementation reads each image using OpenCV and converts the OpenCV BGR representation into RGB format for visualization and further processing. Each image is resized to:

512 × 512 pixels

before the noise-generation and filtering operations.

The notebook demonstrates the processing of the uploaded images and subsequently performs the complete experiment across all uploaded images.

4.2 Reason for Using Artificial Noise

Instead of searching for separate noisy versions of the same images, artificial noise was added to the original images. This provides a known reference image against which the restored image can be compared.

The original image is treated as the reference, while the noisy and filtered images are treated as degraded and restored versions respectively.

This approach makes it possible to calculate objective image-quality metrics such as MSE, PSNR, and SSIM.

5. PROPOSED EXPERIMENTAL DESIGN

Instead of treating the project as a single processing pipeline, the experiment was organized into two independent noise conditions.

Experiment A — Gaussian Noise

For each original image:

Original Image → Gaussian Noise → Three Spatial Filters → Evaluation

The three filtering branches are:

- Mean Filter
- Median Filter

- Gaussian Filter

Experiment B — Salt-and-Pepper Noise

For each original image:

Original Image → Salt-and-Pepper Noise → Three Spatial Filters → Evaluation

Again, the three filters are:

- Mean Filter
- Median Filter
- Gaussian Filter

The results from all three images are collected into a common results table. The average MSE, PSNR, and SSIM are then calculated separately for each combination of noise type and filter.

This experimental design allows the filtering techniques to be compared under identical image-processing conditions.

6. METHODOLOGY

6.1 Stage 1: Image Acquisition

Three input images were uploaded into Google Colab using the file-upload facility.

Each image was read using OpenCV.

Input Images

```
|
|— img1.jpeg
|— img2.jpeg
└— img3.jpeg
```

6.2 Stage 2: Image Standardization

The uploaded images may have different dimensions. To ensure consistent processing, each image was resized to:

$512 \times 512 \times 3$

where:

- 512 = image height
- 512 = image width
- 3 = RGB color channels

The images were also converted from OpenCV's BGR representation to RGB representation.

6.3 Stage 3: Controlled Noise Generation

Two types of artificial noise were introduced.

A. Gaussian Noise

Gaussian noise was generated using a normal distribution.

The implementation used:

- Mean = 0
- Standard deviation = 25

The noisy pixel values were clipped to the valid range of 0–255.

The general noise model can be represented as:

$$\begin{aligned} &[\\ g(x,y) &= f(x,y) + n(x,y) \\ &] \end{aligned}$$

where:

- $(f(x,y))$ is the original image,
- $(n(x,y))$ is Gaussian noise,
- $(g(x,y))$ is the noisy image.

B. Salt-and-Pepper Noise

Salt-and-pepper noise was introduced by randomly changing selected pixels to extreme intensity values.

- Salt pixels were assigned a value of 255.

- Pepper pixels were assigned a value of 0.
- The noise amount used in the experiment was 0.05.

This produces isolated bright and dark pixels throughout the image.

7. SPATIAL FILTERING TECHNIQUES

Three filters were selected for comparison.

7.1 Mean Filter

The Mean Filter replaces a pixel with the average of the pixels within a specified neighborhood.

A 5×5 kernel was used in the implementation.

Conceptually:

$$\begin{aligned} &[\\ g(x,y) &= \frac{1}{N} \sum_{(i,j) \in S} f(i,j) \\ &] \end{aligned}$$

where SS represents the local neighborhood.

Mean filtering can reduce random variations, but because it averages neighboring pixels, it can also blur edges and fine details.

7.2 Median Filter

The Median Filter replaces a pixel with the median value of its neighborhood.

A kernel size of 5 was used.

Unlike averaging, median filtering is based on the ordered values within the neighborhood. This makes it particularly useful for removing isolated extreme pixel values.

Therefore, its behavior against salt-and-pepper noise was specifically investigated in this project.

7.3 Gaussian Filter

The Gaussian Filter performs weighted smoothing using a Gaussian-shaped kernel.

A 5×5 kernel was used, with the standard deviation selected automatically by OpenCV.

The central pixels receive greater weight than pixels farther from the center.

Gaussian filtering is commonly used for reducing smooth/random variations while maintaining a more natural form of image smoothing compared with simple averaging.

8. EXPERIMENTAL COMPARISON

For every input image, six restoration conditions were evaluated:

Noise Type	Filter
Gaussian Noise	Mean Filter
Gaussian Noise	Median Filter
Gaussian Noise	Gaussian Filter
Salt-and-Pepper Noise	Mean Filter
Salt-and-Pepper Noise	Median Filter
Salt-and-Pepper Noise	Gaussian Filter

For each condition, the restored image was compared with the corresponding original clean image.

The notebook automatically collected the results for all uploaded images and calculated the average values for each noise-filter combination.

9. PERFORMANCE EVALUATION

Three evaluation metrics were used.

9.1 Mean Squared Error (MSE)

MSE measures the average squared difference between the original and restored image pixels.

$$[\text{MSE} = \frac{1}{N} \sum_{i=1}^N (I_i - \hat{I}_i)^2]$$

where:

- (I_i) = original pixel value

- $\hat{I}_i$ = restored pixel value
- N = total number of pixels

A lower MSE indicates a smaller numerical difference between the original and restored images.

9.2 Peak Signal-to-Noise Ratio (PSNR)

PSNR is derived from MSE and is expressed in decibels.

$$PSNR = 10 \log_{10} \left(\frac{MAX_I^2}{MSE} \right)$$

For 8-bit images:

$$MAX_I = 255$$

Higher PSNR generally indicates lower reconstruction error.

9.3 Structural Similarity Index (SSIM)

SSIM evaluates structural similarity between two images. Unlike MSE, which mainly considers pixel-wise differences, SSIM considers structural information.

The SSIM value is generally interpreted such that a value closer to 1 indicates greater structural similarity.

In this project, SSIM was calculated using the `skimage.metrics.structural_similarity` implementation with the RGB channels handled using the `channel_axis` parameter.

10. IMPLEMENTATION ENVIRONMENT

The project was implemented using Google Colab.

Software and libraries

Component	Used
Programming Language	Python

Development Environment	Google Colab
Image Processing	OpenCV
Numerical Processing	NumPy
Data Handling	Pandas
Visualization	Matplotlib
Image Quality Evaluation	scikit-image

The notebook imports OpenCV, NumPy, Matplotlib, Pandas, and the SSIM and PSNR functions from scikit-image.

11. RESULTS

The complete experiment was performed on the three uploaded images. The results below represent the average values calculated across the images.

11.1 Average Performance Results

NOISE TYPE	FILTER	MSE	PSNR (DB)	SSIM
GAUSSIAN NOISE	Gaussian Filter	82.54	29.03	0.7102
GAUSSIAN NOISE	Mean Filter	91.80	28.65	0.7505
GAUSSIAN NOISE	Median Filter	98.08	28.34	0.6844
SALT-AND-PEPPER NOISE	Median Filter	57.55	30.88	0.8937
SALT-AND-PEPPER NOISE	Mean Filter	124.92	27.24	0.6635
SALT-AND-PEPPER NOISE	Gaussian Filter	136.12	26.83	0.6135

These are the average values generated by the submitted Colab implementation.

12. RESULT ANALYSIS

12.1 Gaussian Noise

For Gaussian noise, the Gaussian Filter produced:

- MSE = 82.54

- PSNR = 29.03 dB
- SSIM = 0.7102

The Mean Filter produced:

- MSE = 91.80
- PSNR = 28.65 dB
- SSIM = 0.7505

The Median Filter produced:

- MSE = 98.08
- PSNR = 28.34 dB
- SSIM = 0.6844

Based on MSE and PSNR, the Gaussian Filter gave the strongest numerical restoration result for the Gaussian-noise experiment. It had the lowest MSE and highest PSNR among the three filters.

However, SSIM shows a slightly different behavior: the Mean Filter achieved an SSIM of 0.7505, compared with 0.7102 for the Gaussian Filter. This indicates that the filter producing the lowest pixel-level error did not simultaneously produce the highest structural-similarity score in this experiment.

This difference is important because it demonstrates that MSE/PSNR and SSIM measure different aspects of image quality.

12.2 Salt-and-Pepper Noise

The results for salt-and-pepper noise were considerably different.

The Median Filter produced:

- MSE = 57.55
- PSNR = 30.88 dB
- SSIM = 0.8937

The Mean Filter produced:

- MSE = 124.92

- PSNR = 27.24 dB
- SSIM = 0.6635

The Gaussian Filter produced:

- MSE = 136.12
- PSNR = 26.83 dB
- SSIM = 0.6135

The Median Filter therefore showed a substantially smaller numerical error and higher PSNR and SSIM for salt-and-pepper noise compared with the other two filters.

This behavior is consistent with the nature of salt-and-pepper noise. Since this noise introduces isolated extreme-valued pixels, a median operation can suppress these outliers without relying on their extreme intensity values in the same way that averaging-based filters do.

13. VISUAL RESULTS

The notebook contains visual comparisons of the original, noisy, and filtered images.

The following figures should be inserted into this section from the outputs of your Colab notebook.

Original Image



Figure 1: Original input image before noise addition.

Original Image



Image with Gaussian Noise

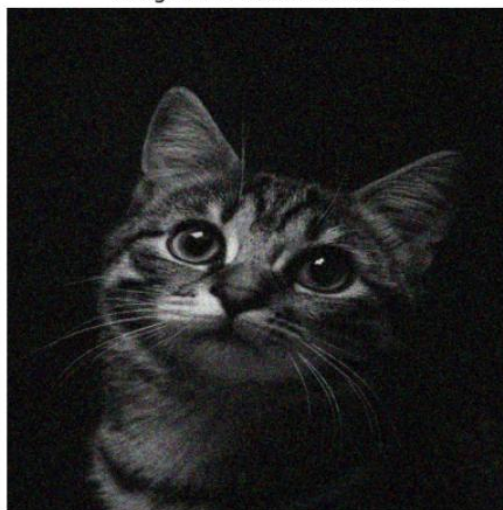


Figure 2: Original image and image degraded using artificially generated Gaussian noise.

Original Image



Salt-and-Pepper Noise

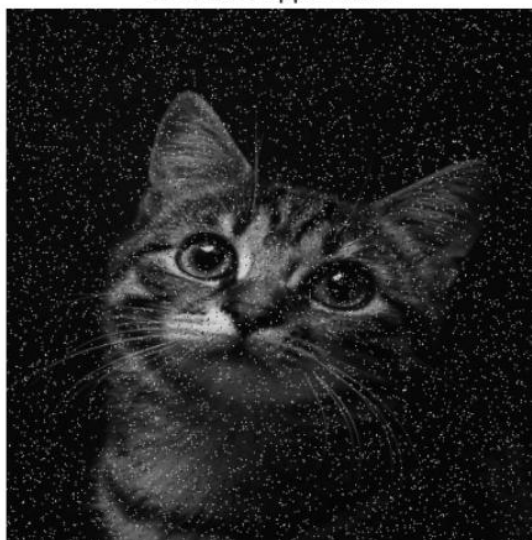


Figure 3: Original image and image degraded using salt-and-pepper noise.

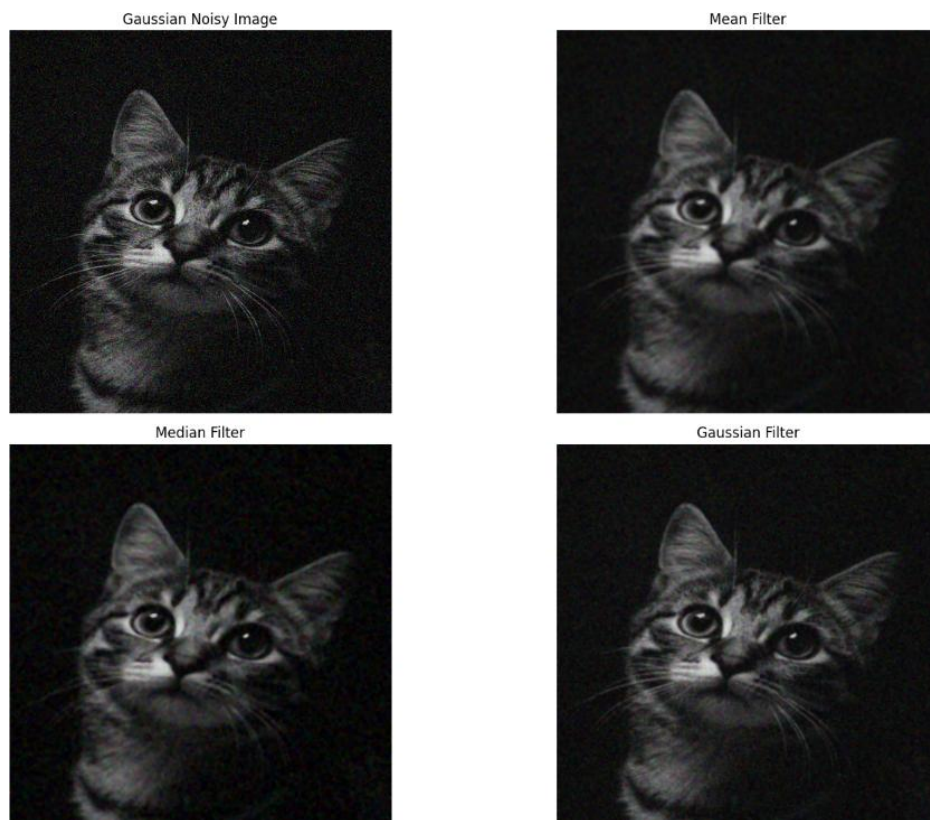


Figure 4: Comparison of spatial filters applied to an image affected by Gaussian noise.

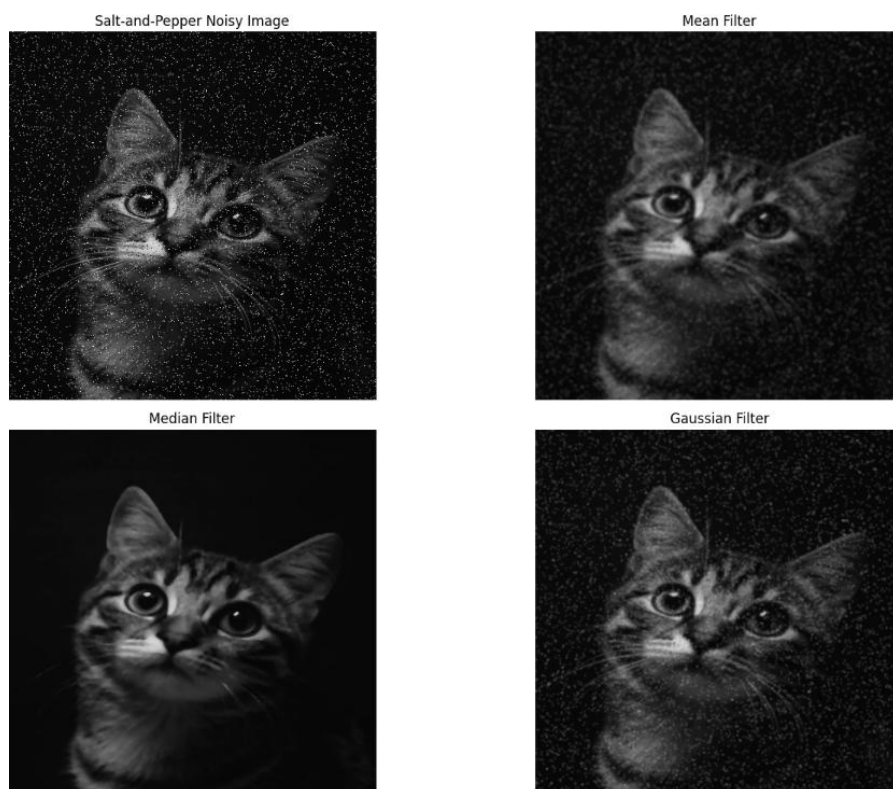


Figure 5: Comparison of spatial filters applied to an image affected by salt-and-pepper noise.

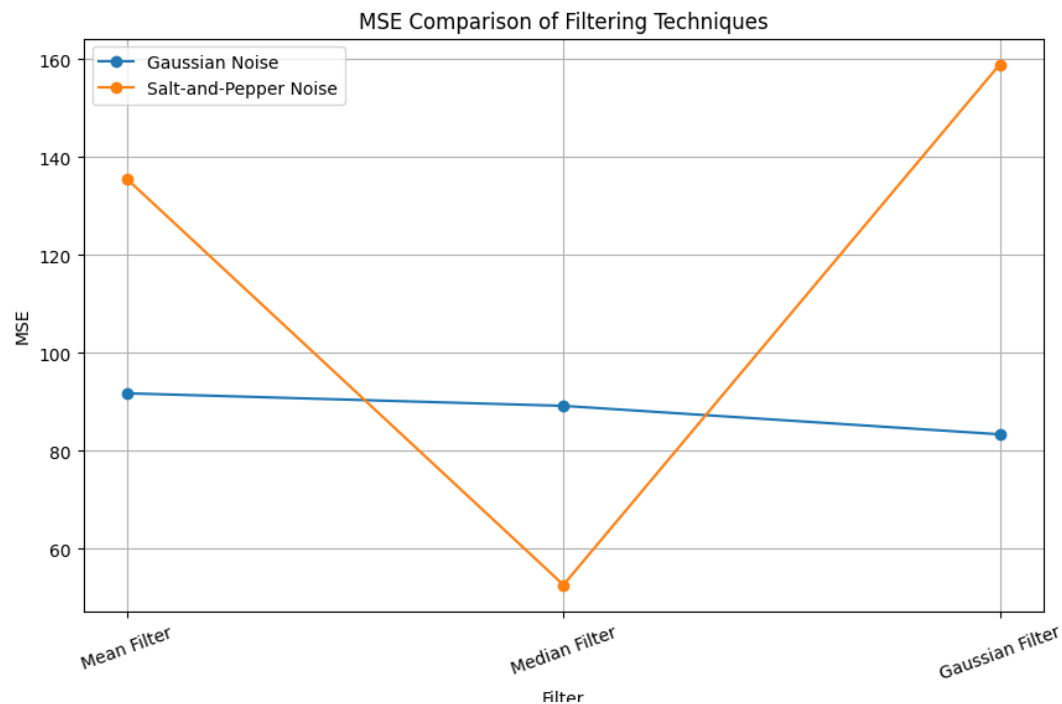
14. GRAPHICAL COMPARISON

Three graphical comparisons were generated in the notebook.

14.1 MSE Comparison

The MSE graph compares the average numerical reconstruction error for the three filters under both noise conditions.

The results show that the average MSE for salt-and-pepper noise was lowest with the Median Filter at 57.55, whereas the Gaussian Filter produced the lowest MSE for Gaussian noise at 82.54.



14.2 PSNR Comparison

The PSNR graph provides a complementary view of restoration quality.

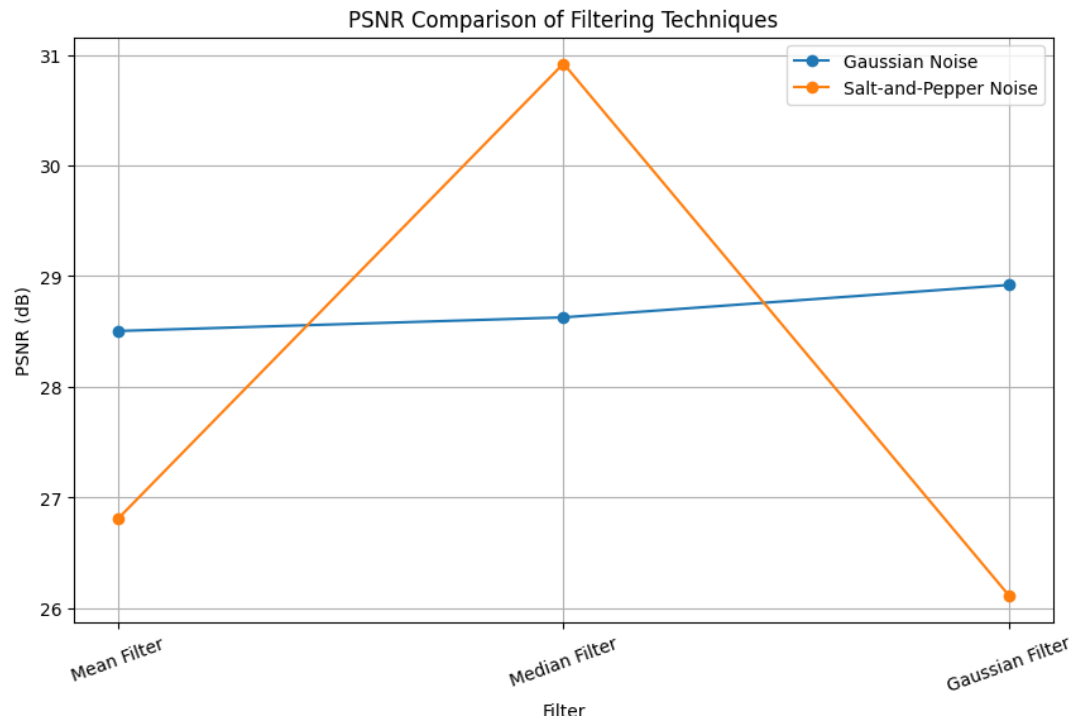
For Gaussian noise:

Gaussian Filter → 29.03 dB

For salt-and-pepper noise:

Median Filter → 30.88 dB

The higher PSNR values correspond to the lower MSE values observed in the respective experiments.



14.3 SSIM Comparison

The SSIM results provide information about structural similarity.

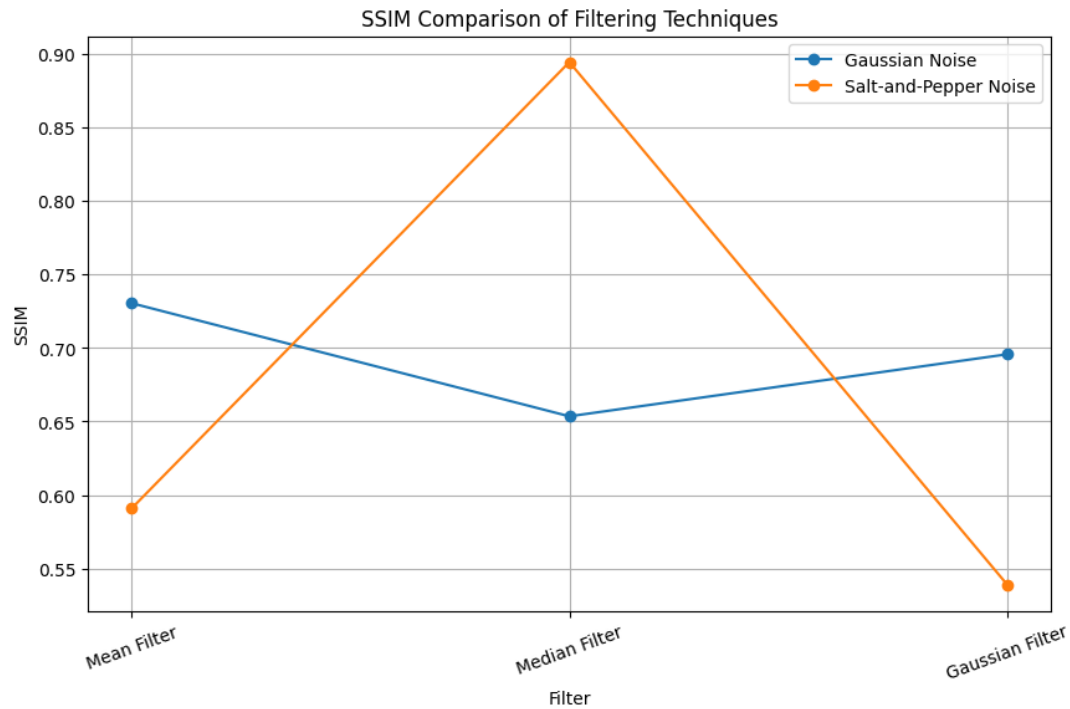
For Gaussian noise, the Mean Filter produced the highest SSIM:

0.7505

For salt-and-pepper noise, the Median Filter produced the highest SSIM:

0.8937

This demonstrates that the different evaluation measures can provide complementary information about restoration quality.



15. DISCUSSION

The experiment demonstrates that spatial filtering performance is strongly related to the characteristics of the noise.

For Gaussian noise, the Gaussian Filter produced the lowest MSE and highest PSNR among the tested filters. This suggests that Gaussian smoothing was effective at reducing the random intensity variations introduced by the Gaussian noise model.

The Mean Filter produced a higher SSIM than the Gaussian Filter in the same experiment. Therefore, the numerical results should not be interpreted using only one metric. MSE and PSNR primarily describe pixel-level reconstruction error, while SSIM provides information about structural similarity.

The salt-and-pepper experiment produced a clearer difference between the filters. The Median Filter achieved an MSE of 57.55, PSNR of 30.88 dB, and SSIM of 0.8937. These values were better than those obtained with the Mean and Gaussian filters for this noise condition.

The experiment therefore illustrates an important principle of image processing:

The effectiveness of a denoising method depends on the statistical characteristics of the noise.

A filter should therefore be selected according to the type of degradation present in the image.

16. LIMITATIONS

Although the project demonstrates the fundamental principles of image denoising, several limitations should be considered.

16.1 Small Dataset

Only three images were used in the experiment. Therefore, the average results should be interpreted as observations from this selected dataset rather than as universal performance values.

16.2 Artificial Noise

The noise was generated artificially using predefined parameters. Real-world image noise can have more complex characteristics.

16.3 Fixed Filter Parameters

The experiment uses fixed kernel sizes:

- 5×5 Mean Filter
- 5×5 Median Filter
- 5×5 Gaussian Filter

Different kernel sizes may produce different results.

16.4 Limited Noise Models

Only Gaussian noise and salt-and-pepper noise were considered. Other forms of noise, such as speckle noise, Poisson noise, or periodic noise, were not investigated.

16.5 Limited Filtering Methods

Only three spatial filters were compared. More advanced restoration techniques were outside the scope of this project.

17. FUTURE SCOPE

The project can be extended in several ways.

1. Increase the dataset size to include a larger and more diverse collection of images.
2. Test multiple noise intensities instead of using a single noise level.
3. Compare different kernel sizes such as 3×3 , 5×5 , and 7×7 .

4. Include adaptive median filtering.
5. Investigate additional noise models such as speckle and Poisson noise.
6. Compare spatial filtering with frequency-domain filtering.
7. Investigate edge-preserving filters.
8. Compare traditional filters with more advanced image-restoration techniques.
9. Develop a simple graphical user interface where a user can upload an image and select a noise type and filter.
10. Study the effect of denoising on subsequent image-processing tasks such as edge detection and segmentation.

18. CONCLUSION

This project investigated image denoising using three spatial filtering techniques: Mean Filter, Median Filter, and Gaussian Filter. Two types of artificial noise—Gaussian noise and salt-and-pepper noise—were introduced into three original images, and the noisy images were processed using each of the selected filters.

The results demonstrate that filter performance varies according to the type of noise. For Gaussian noise, the Gaussian Filter produced the lowest average MSE of 82.54 and the highest average PSNR of 29.03 dB among the tested filters. The Mean Filter produced the highest SSIM for this noise condition, with a value of 0.7505.

For salt-and-pepper noise, the Median Filter produced the lowest average MSE of 57.55, the highest average PSNR of 30.88 dB, and the highest SSIM of 0.8937 among the tested filters.

The experiment therefore demonstrates that no single filtering technique should be assumed to be optimal for every type of image noise. The characteristics of the degradation should be considered when selecting an appropriate restoration method.

Overall, the project successfully demonstrates the use of fundamental Digital Image Processing concepts including artificial noise generation, spatial filtering, image restoration, quantitative evaluation, visualization, and comparative analysis.